ML and Survival Models on Pediatric Brain Tumor

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Introduction

Pediatric brain tumors are the most common solid tumors in children.

They are a leading cause of cancer-related mortality in the pediatric population

Survival outcomes vary greatly due to tumor heterogeneity, treatment complexity, and healthcare resources.

This study analyzes electronic health records (EHRs) from a national referral center (Serbia, 2007–2016).

Identify key clinical and pathological factors affecting survival

Dataset Description

Source: National Oncology and Radiology Institute, Serbia
Period: 2007–2016
Sample size: 174 pediatric patients with primary brain tumors
Study type: Retrospective cohort
Number of variables: 34

Demographics
sex, year of birth

Clinical presentation
neurological deficits, hormonal abnormalities and other

Tumor characteristics
location, histology, disease extent

Treatments
surgery, radiotherapy, chemotherapy

Outcomes
vital status, overall survival time in months

Data Cleaning

Renamed variables for code compatibility (e.g., Sex, Year_of_birth)

Missing data handling:

- Removed 1 record with many missing values
- Logical imputation for radiotherapy/chemotherapy doses, operation_type
- Multiple imputation for conditional fields (e.g., Concomitant_HT)

Feature engineering:

- Aggregated rare histologies (<10 obs)
- Created binary Operation_type_bin (GTR vs. other)
- Combined CSF cytology indicators into one field
- Transform categorical variables in dummy
- Derived target variable Death from Status_OS

Exploratory Data Analysis

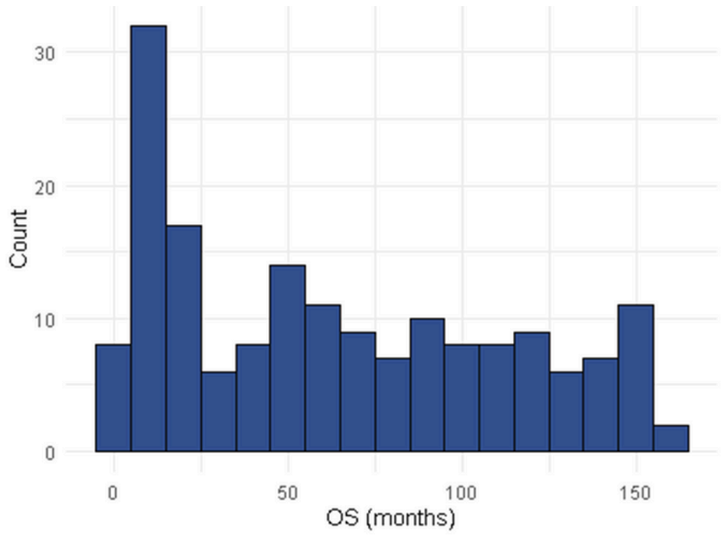
Survival status:

- Alive at last follow-up: 94 patients
- Deceased: 79 patients

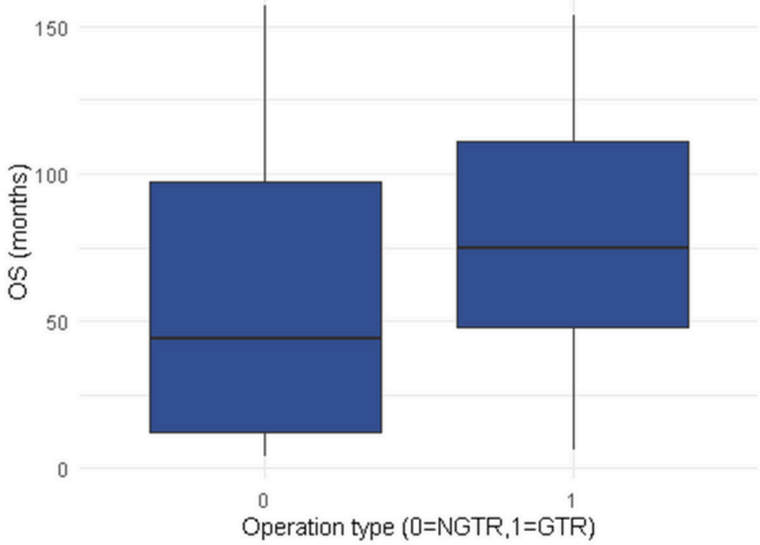
Gender distribution:

- Male: 90
- Female: 83

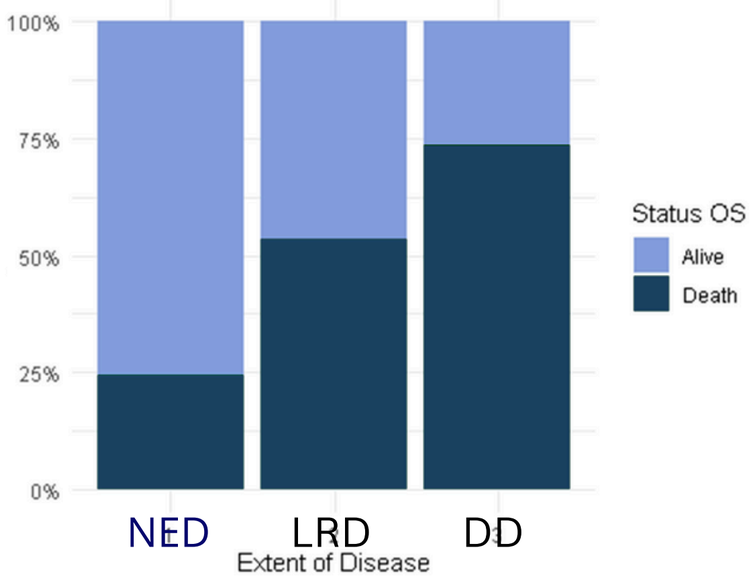
Overall surival distribution



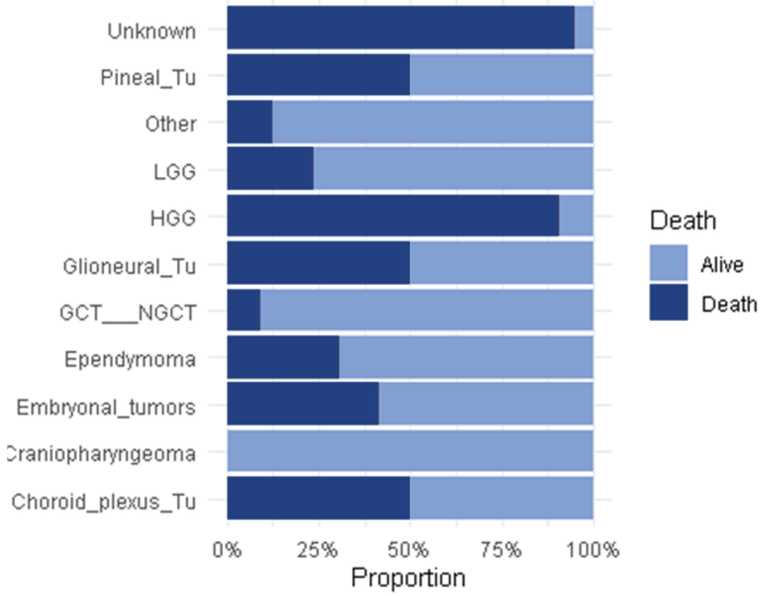
Overall survival by Operation_bin



Mortality by Extent of Disease



Mortality by Histotype



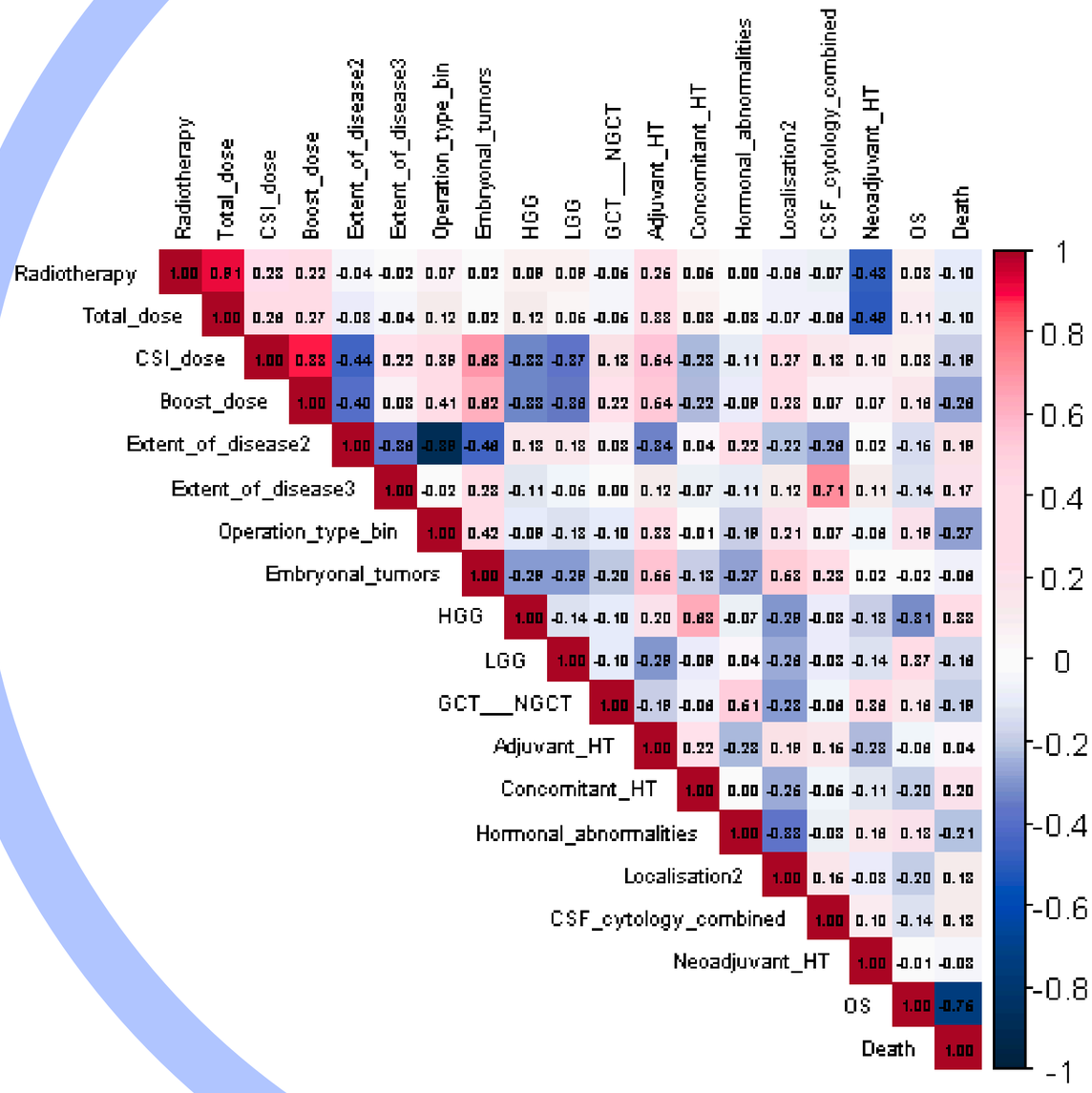
Correlation analysis

Expected Correlations

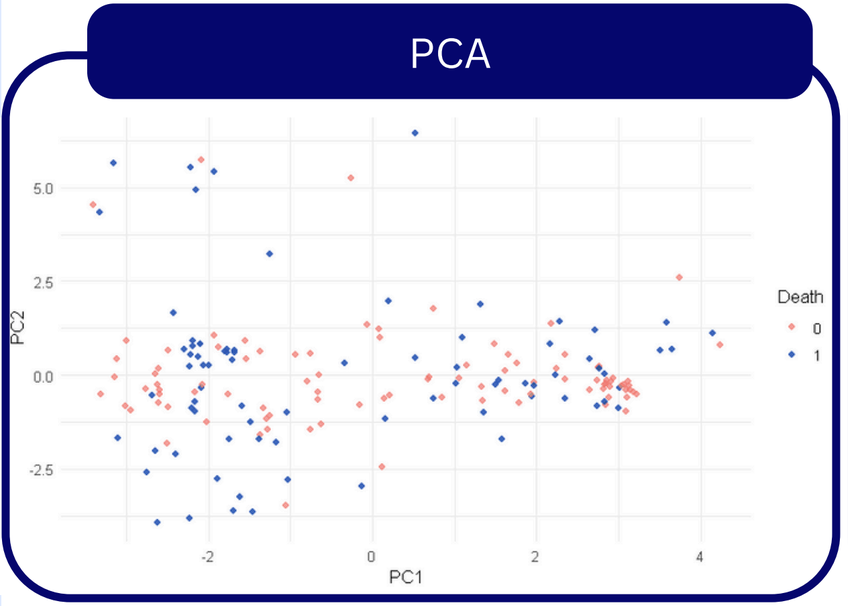
- Radiotherapy ~ Total Dose, Death ~ OS

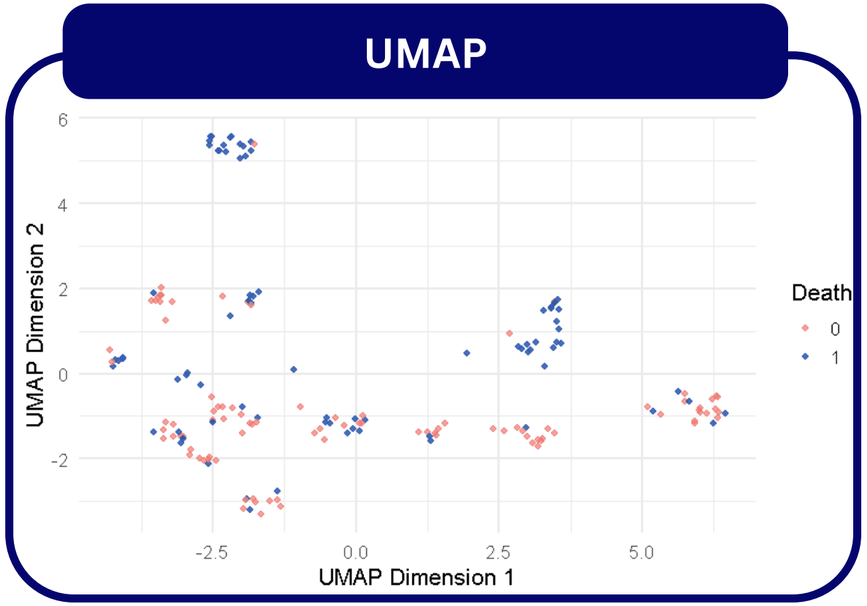
Relevant and Non-Trivial

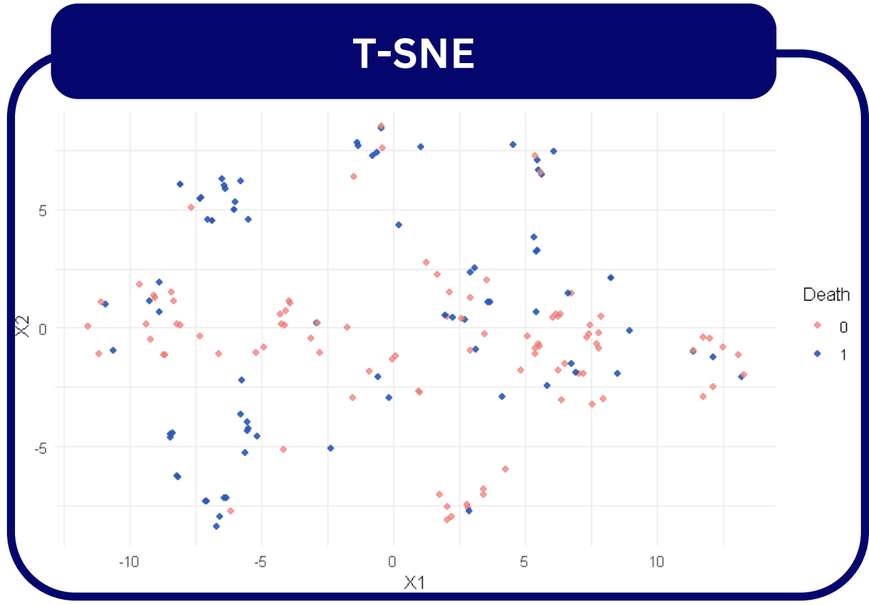
- Extent 3 ~ CSF Cytology → r = 0.71
  - Disseminated disease aligns with CSF positivity.
- HGG ~ Concomitant Chemotherapy → r = 0.63
  - Aggressive gliomas often treated with concurrent chemoradiotherapy.
- Embryonal Tumors ~ CSI Dose → r = 0.68
  - Embryonal tumors commonly receive craniospinal irradiation.
- Extent 2 ~ Gross Total Resection (GTR) → r = −0.89
  - Local residual disease is rarely fully resected.



Dimensioanality Reduction







Dimensionality reduction techniques highlight the complexity of the data:

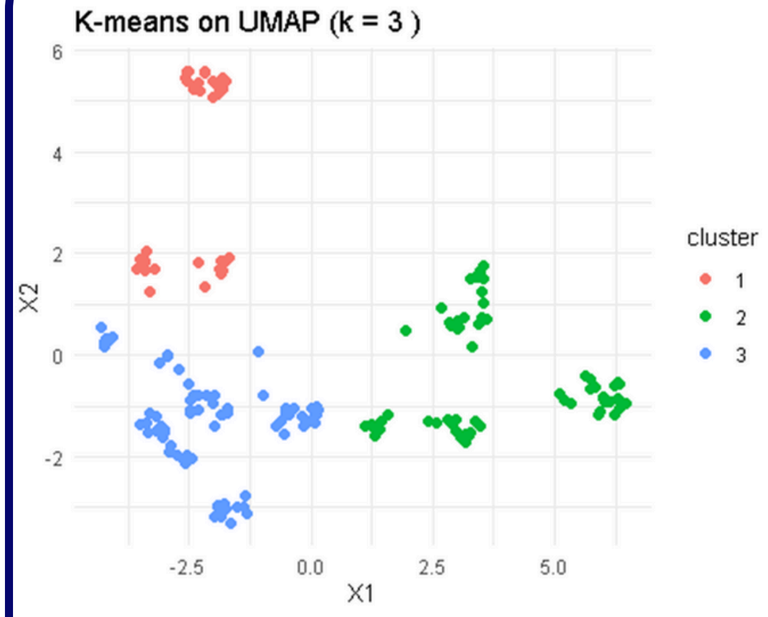
- no technique clearly separated survivors from non-survivors
- PCA failed to separate groups, explained variance was low, it captured less than 30% of the total variance in the first two components
- UMAP and t-SNE suggest subtle non-linear structures.

Unsupervised Learning

- K-means on raw data
- Hierarchical clustering
- K-means on UMAP

no clustering on raw data, best result with K=3 on UMAP

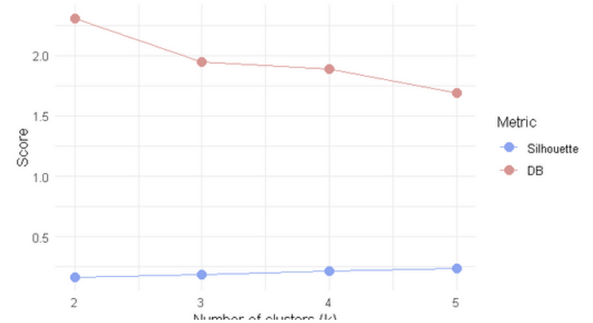
Result clustering K=3 +UMAP

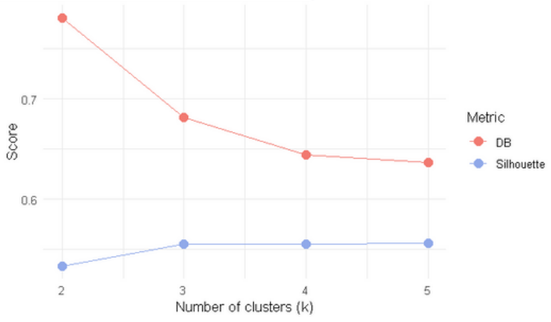


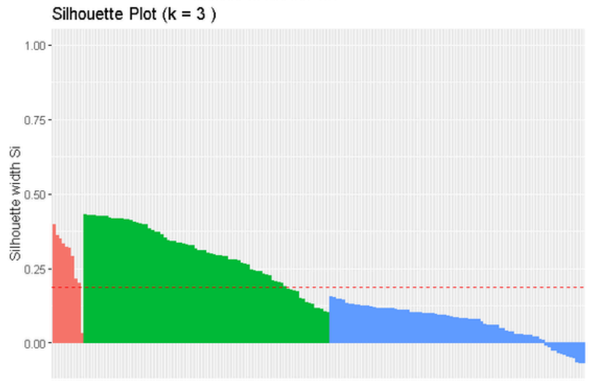
Silhouette :0.56

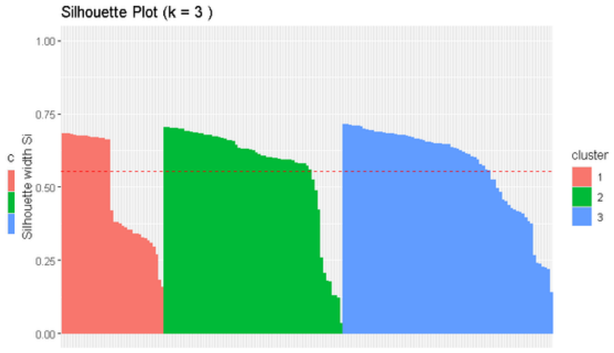
DB :0.68

Metrics raw vs UMAP









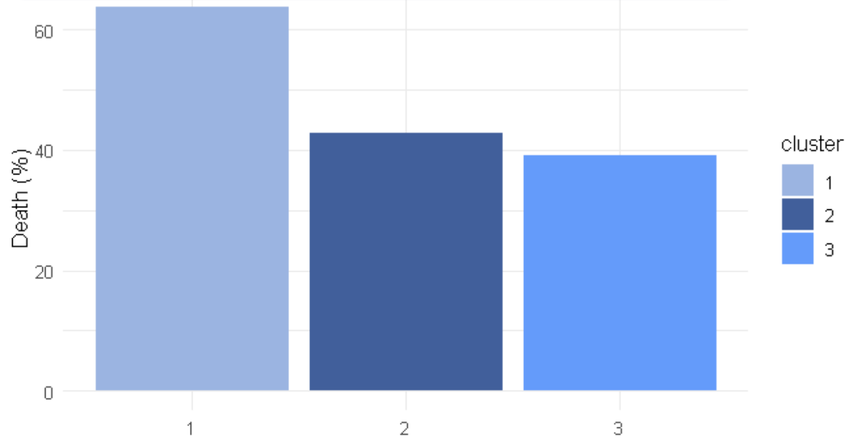
Cluster Profiles

C1-High-risk,Poor Prognosis

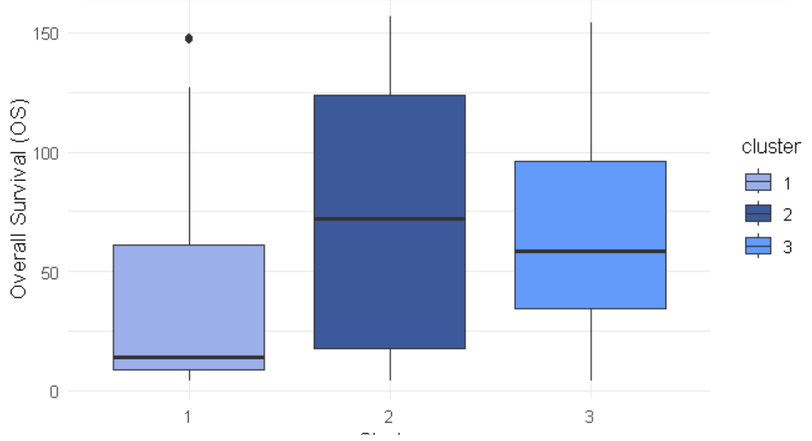
C2- Intermediate-risk, Heterogeneous Outcomes

C3- Aggressive Tumors, Good Management

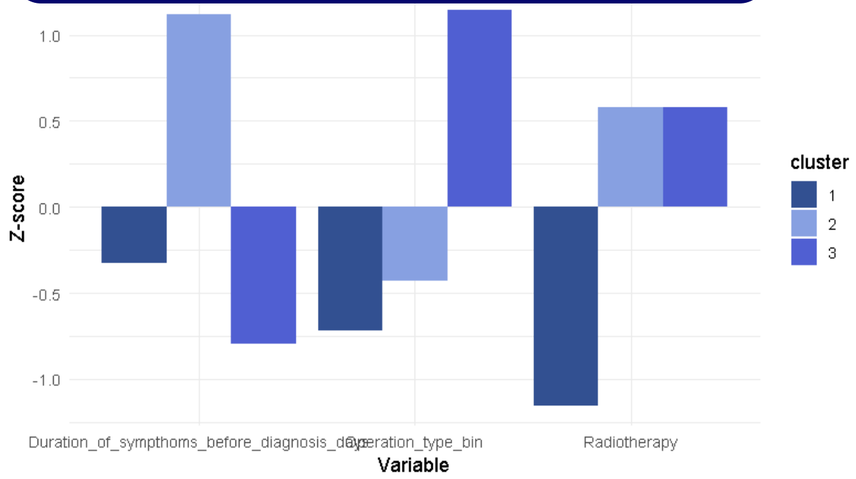
Proportion of Death



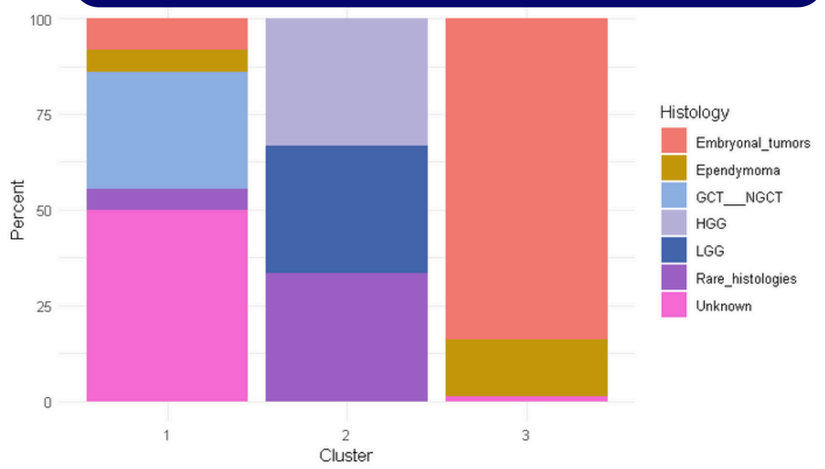
Overall survival



Dur.synt, operation, radioteraphy



Histology composition

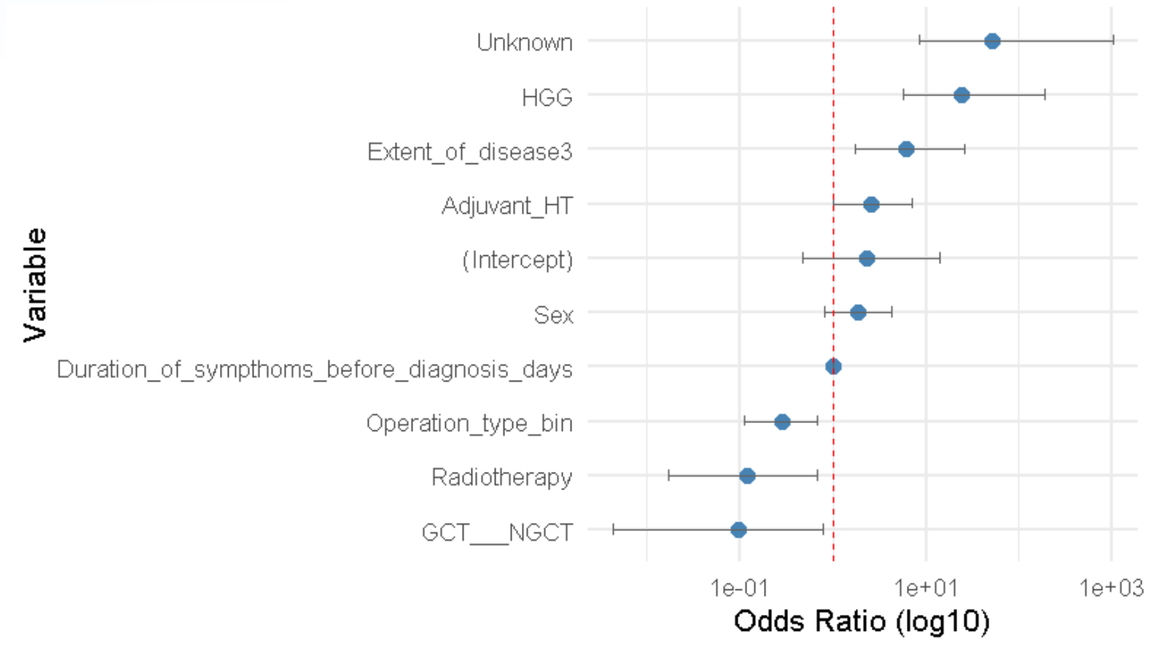


Odds Ratio– GLM

Performed a stepwise logistic regression (both directions) based on AIC to identify the most relevant clinical predictors of mortality (Death).
The model shows good discriminative ability (AIC = 174.16).

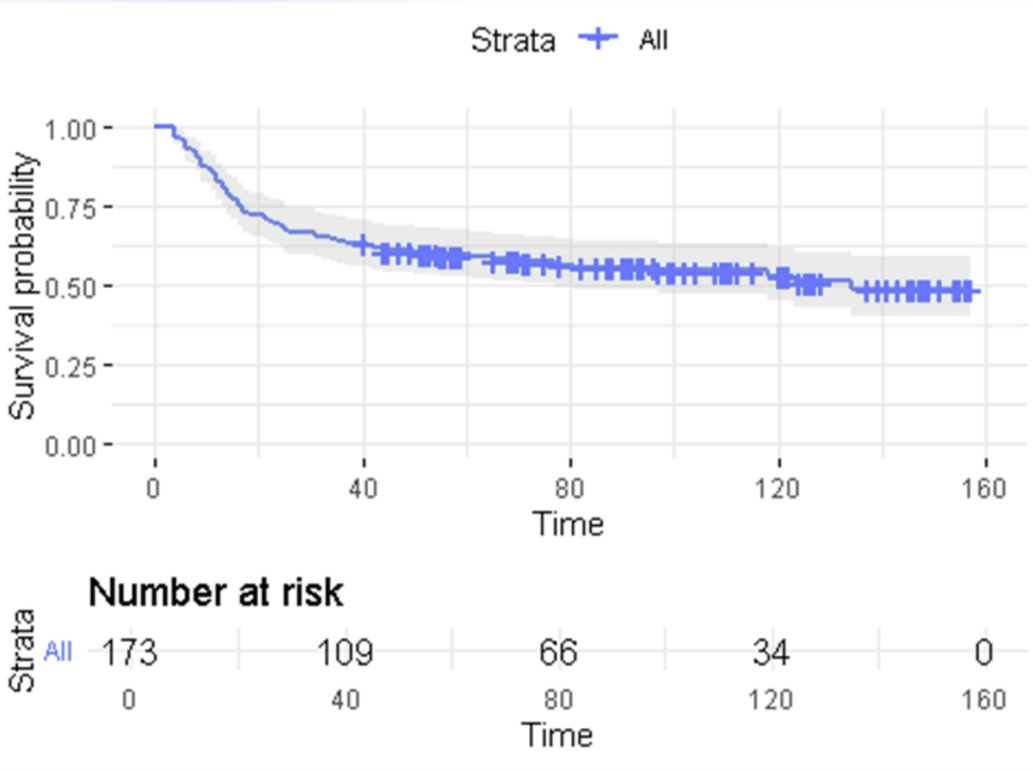
Variabile	***OR***	***95% CI***	*p-value*	***Interpretation***
HGG	23,81	5.61 – 187.75	<0.001	Greatly increased risk of death
Unknown	52,14	8.57 – 1023.31	<0.001	Strong association with mortality
Radiotherapy	0,12	0.017 – 0.669	0,0019	Protective effect
Operation_type_bin	0,28	0.11 – 0.68	0,0060	Radical surgery reduces risk
Extent_of_disease3	6,10	1.74 – 25.79	0,0070	Diffuse disease extent increases risk
Adjuvant_HT	2,60	1.02 – 7.03	*0,0510*	Marginally associated with increased risk
GCT___NGCT	0,098	0.004 – 0.78	*0,0600*	Potentially protective histology

Odds Ratio (OR) with 95% CI (log scale)

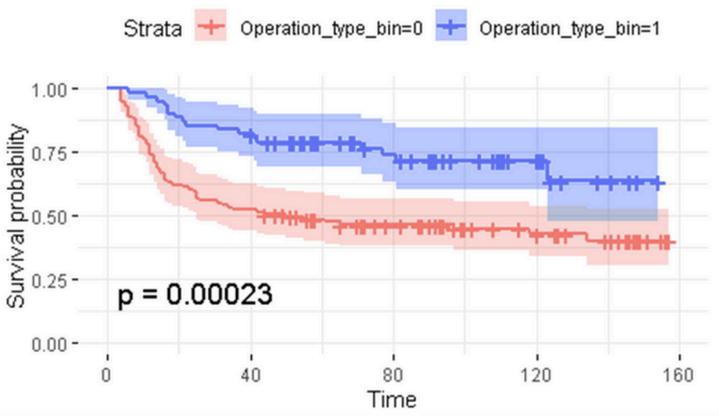


Kaplan Meier Curves

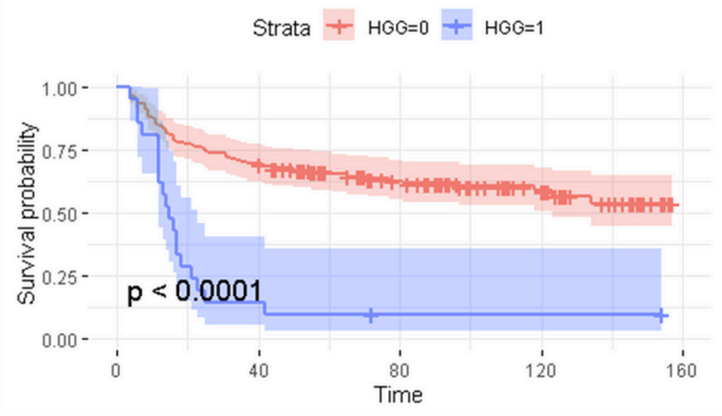
Kaplan-Meier curve - Overall Survival



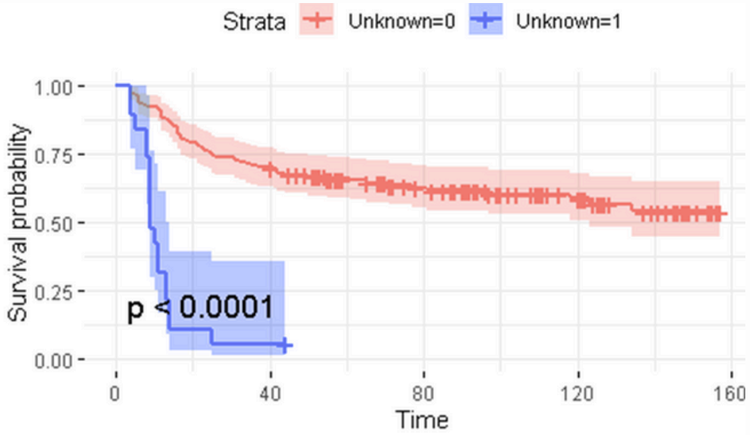
KM - Operation type



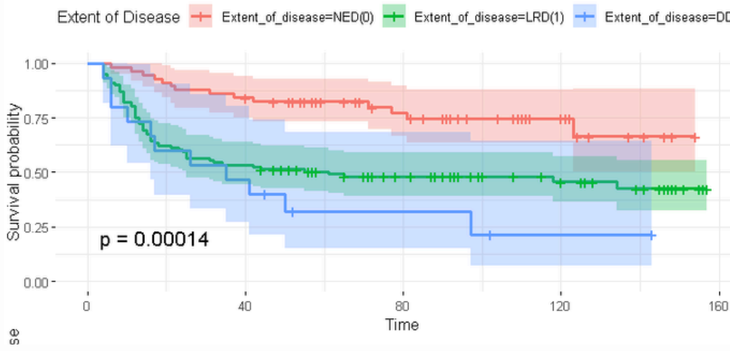
KM - HGG



KM - Unknown histologies



KM - Extent of disease



Cox Model

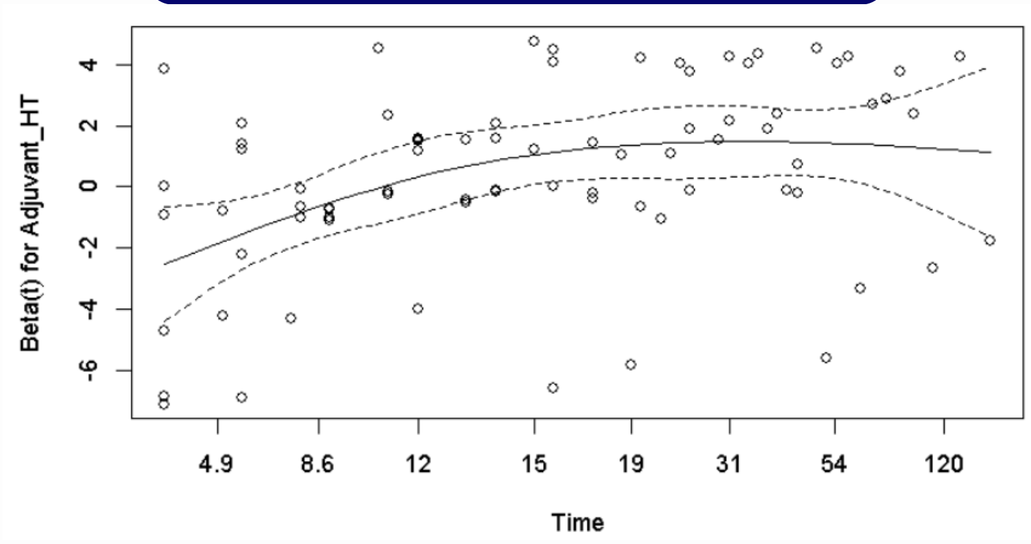
We first fitted a multivariate Cox model using 9 clinically relevant predictors (Sex, HGG, Unknown, Radiotherapy, Extent_of_disease3, Adjuvant_HT...

Good performance:
C-index = 0.813

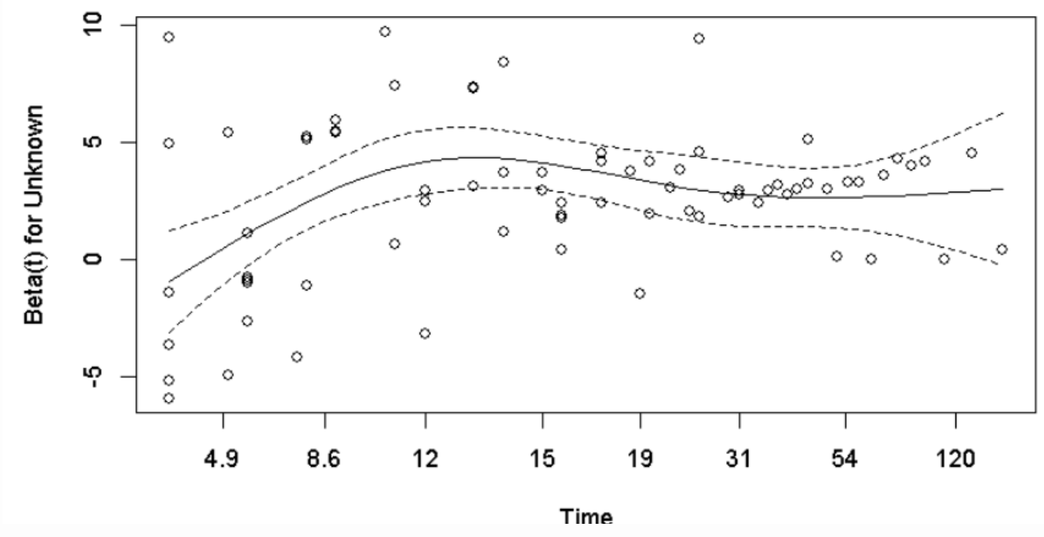
However, the proportional hazards assumption was violated, particularly for:

- Adjuvant_HT
- Unknown histology

Residual for Adjuvant_HT



Residual for Unknown



Deviation from horizontal line suggests time-varying effects → PH assumption violated

Stratified Cox Model

Stratification on Adjuvant_HT and Unknown resolved assumption violations

Final model showed stable coefficients and robust predictors

Global Schoenfeld test: p = 0.95
C-Index 0.78

Variable	HR	95% CI
High-Grade Glioma (HGG)	**9,18**	**[4.78 – 17.64]**
Disseminated Disease (Extent 3)	**3,37**	**[1.63 – 6.97]**
No Gross Total Resection	**0,30**	**[0.18 – 0.60]**
Radiotherapy	**0,17**	**[0.06 – 0.48]**

Non-significant: Sex, GCT___NGCT, Symptom Duration

Supervised Learning

Multicollinearity Check

- Variance Inflation Factor (VIF) analysis was conducted to assess redundancy.
- Final dataset 23 features

Training & Evaluation

- 10-fold cross-validation on training set
- Performance evaluated on 20% hold-out test set
- Optimal threshold applied (Youden Index) to balance sensitivity and specificity.
  - Youden Index=Sensitivity+Specificity−1

Random Forest showed the best performance (AUC = 0.83)

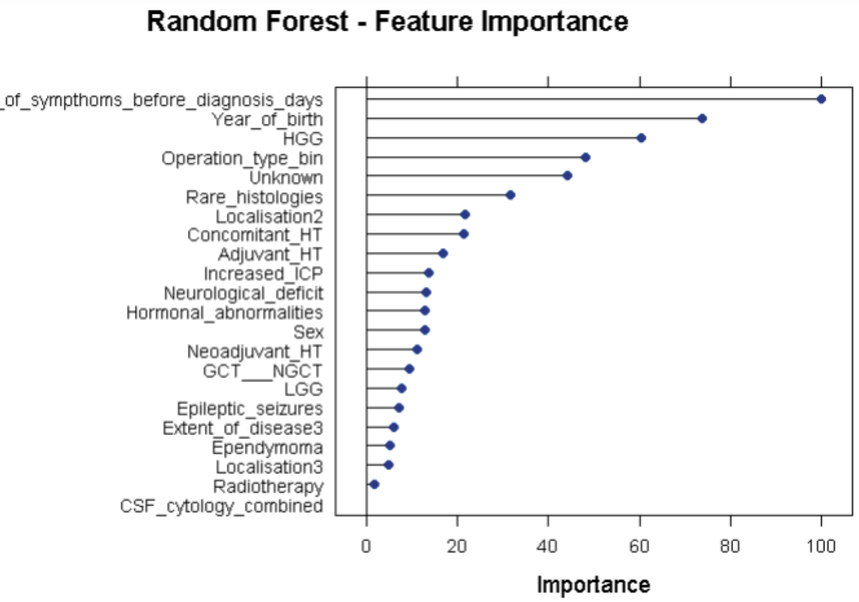
Result with optimal threshold

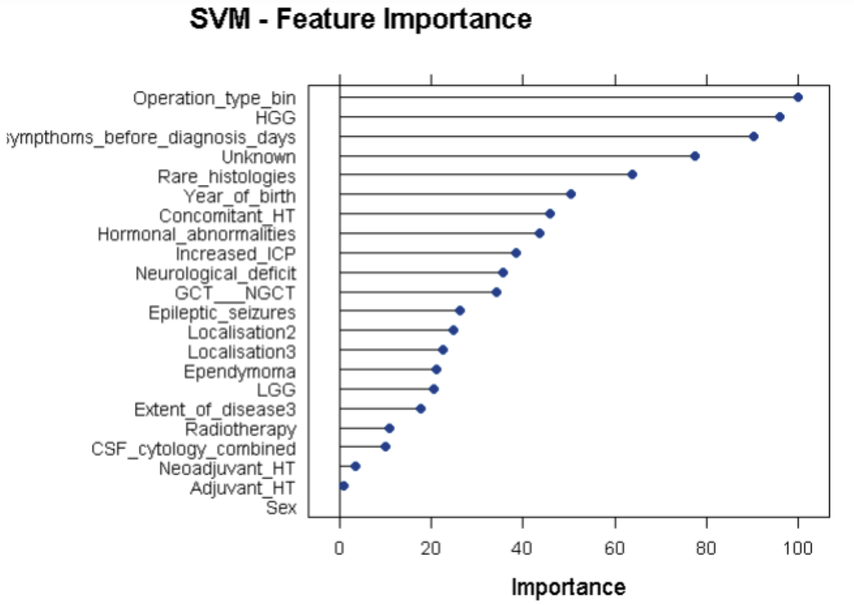
Model	AUC	Accuracy	Sensitivity	Specificity
Random Forest	0,83	0,76	0,64	0,85
XGBoost	0,82	0,76	0,64	0,85
SVM	0,70	0,82	0,79	0,85
Neural Network	0,79	0,76	0,57	0,90
KNN	0,53	0,41	1,00	0,00

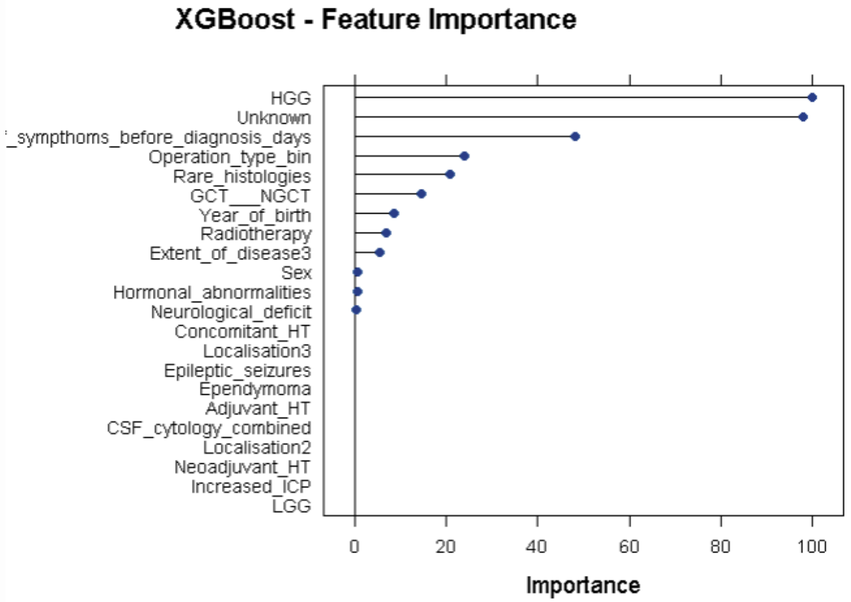
Feature importance

Consistently important features:

- HGG (High-grade glioma)
- Operation_type_bin (GTR)
- Unknown Histology
- Duration of Symptoms







Conclusions

Clinical Findings

Mortality strongly associated with:

- High-grade gliomas (HGG)
- Disseminated disease
- Unknown histology
- No GTR

Radiotherapy and gross total resection were protective

Survival Modeling

Global Cox model: c-index 0.81

Stratified Cox model: C-index 0.78

Unsupervised Learning

No cluster on raw data

UMAP + K-means revealed 3 distinct patient profiles

Clusters showed clinical and prognostic differences

Mortality varied significantly across clusters ($p < 0.05$)

Supervised Learning

Best models: Random Forest , XGBoost, SVM

Top predictors: HGG, unknown histology, surgery type, symptom duration

Limitations

Single-center dataset, moderate sample size
Imputation of missing data
Simplified treatment variables

Thank you for your attention